

22MA201-ENGINEERING MATHEMATICS II

UNIT 2-LP-7- METHODS FOR MODELS FOR LINEAR OSCILLATORS: SIMPLE HARMONIC MOTION

1. SIMPLE MASS SPRING SYSTEM

1.1. Mathematical Formulation of a Vibrating Spring

1.2. Example 1

1.3. Example 2

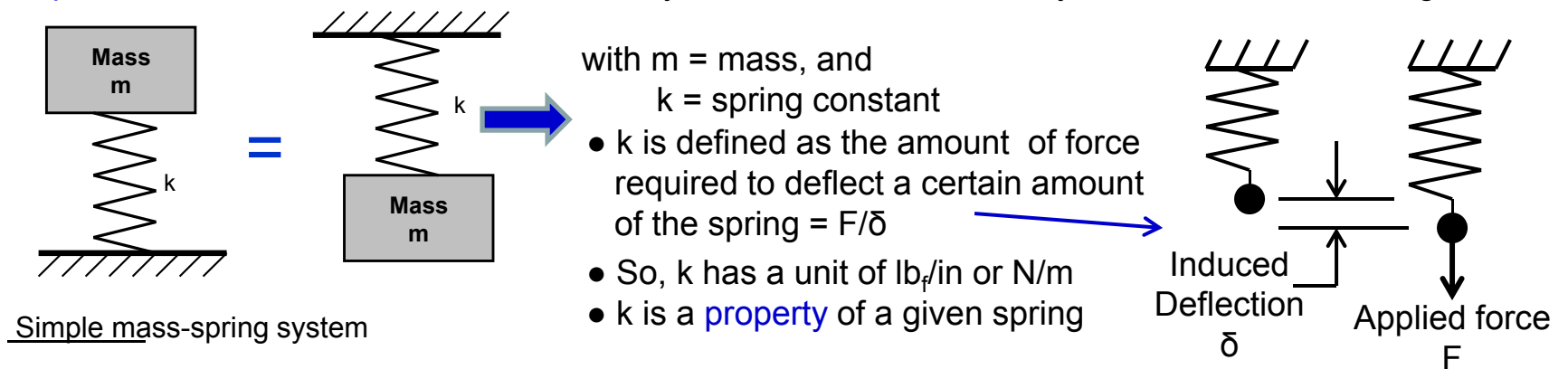
1.4. Example 3

1.5. Discussion Questions

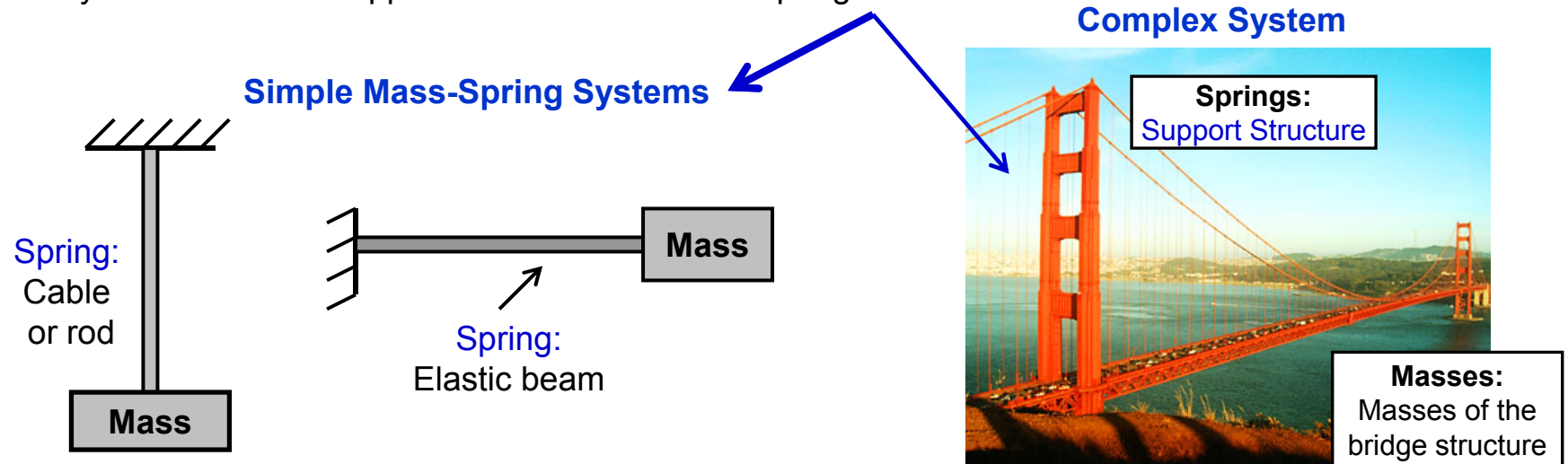
1.1. SIMPLE MASS SPRING SYSTEM

Minimum requirement for Mechanical vibration: a MASS attached to an ELASTIC SUPPORT

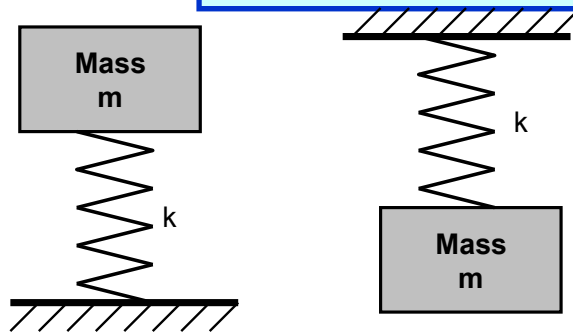
The simplest model for mechanical vibration analysis is a **MASS-SPRING** system as illustrated in Figure



- The spring in this system is to support the mass
- Springs in the system need **not** to be “coil” springs
- Any ELASTIC solid support can be viewed as a “spring”



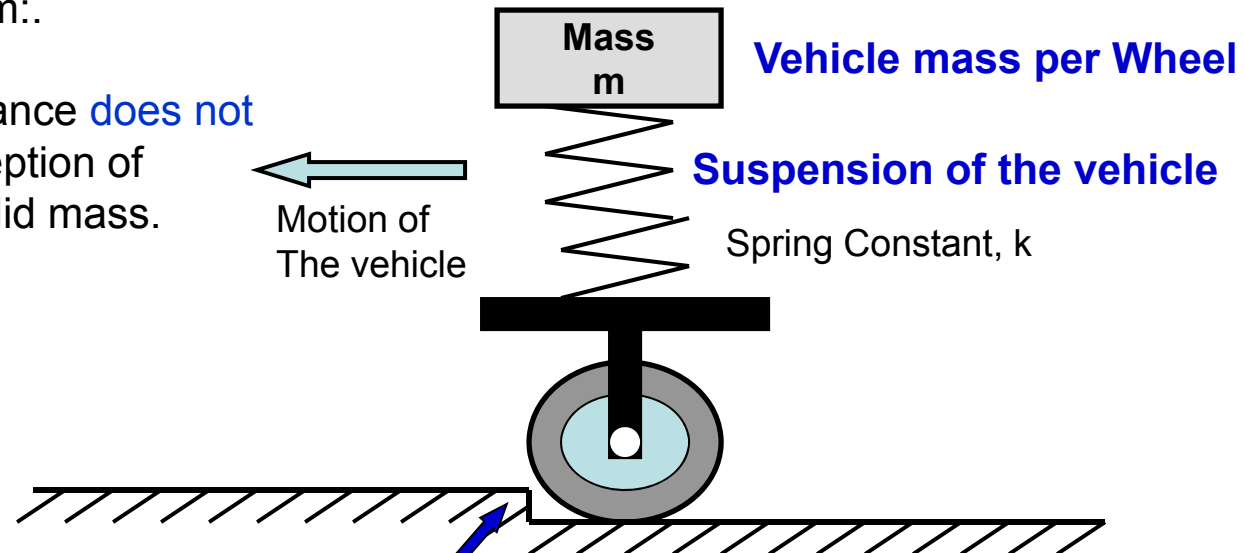
A Case of Simple Mass-Spring Systems in Free Vibration



It takes a **MASS** and **SPRING** (or elastic) support to get the mass to vibrate

The **physical phenomena** of solids in **free vibration** is that the vibration of the solid is induced by an **instantaneous disturbance** either in the form of a force or deformation of the supporting spring, such as illustrated in the vibration of a vehicle induced by its suspension system:.

This initial disturbance **does not exist** after the inception of vibration of the solid mass.

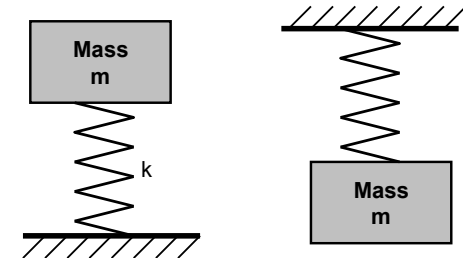


This disturbance (a “blip” on Road surface) causes the mass to begin vibration, and continue to vibrate afterward

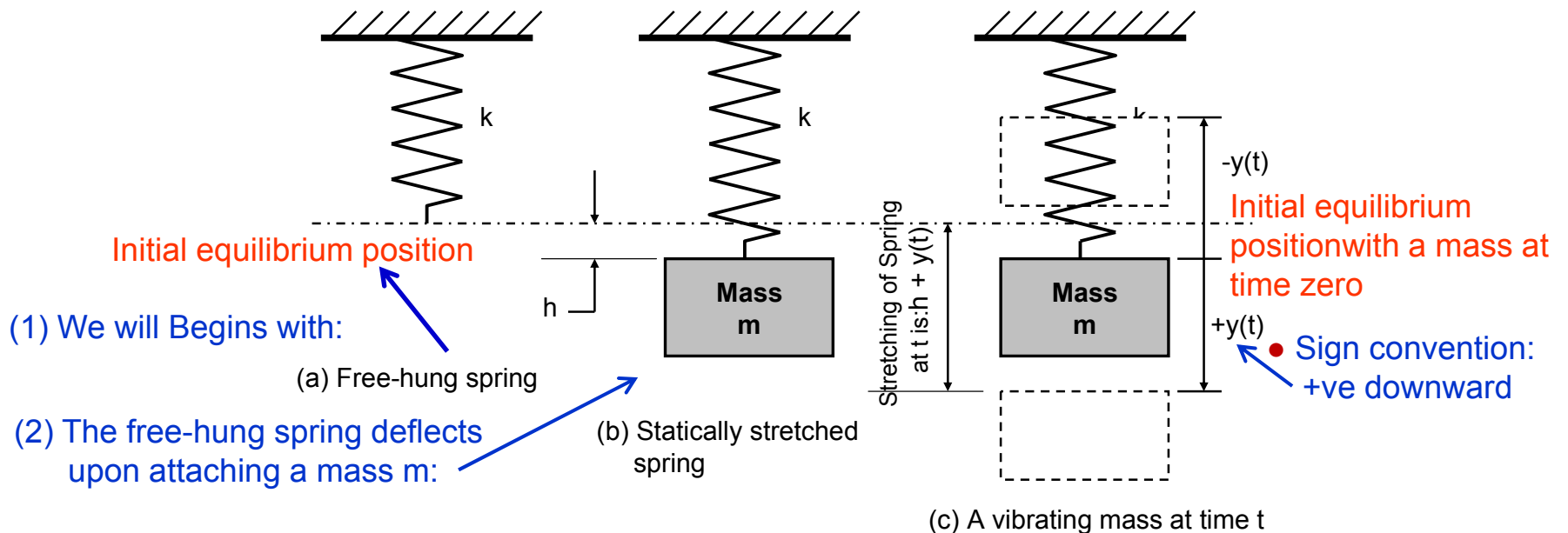
Mathematical Formulation of Simple Mass-Spring Systems in Free Vibration

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for mechanical vibration analysis involves a **MASS** and a **SPRING** (or an elastic support) as illustrated in Figure

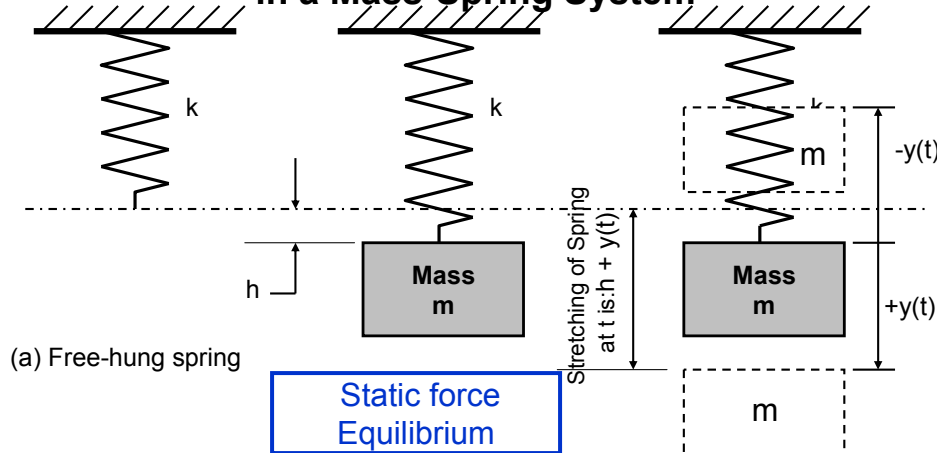


Mathematical Formulation for Free Vibration of a mass:



1.1 Mathematical formulation of a vibrating mass

in a Mass-Spring System

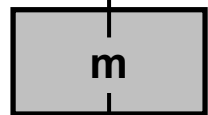


(a) Free-hung spring

Static force
Equilibrium

Dynamic force
Equilibrium at time t

Spring force
 $F_s = kh$



Weight
 $W = mg$

$$+\downarrow \sum F_y = W - F_s = 0$$

$$mg = kh$$

Based on an assumption with **no** air resistance against the motion of the mass and spring. Formulation is on the equilibrium of both static and dynamic forces.

Forces: Weight (**W**); Spring force (**F_s**)

Dynamic force, **F(t)**:

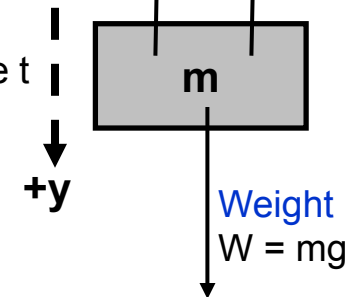
Spring force:

$$F_s = k[h + y(t)]$$

Dynamic (Inertia)
Force, **F(t)**

Displacement
+y(t):

Position at time t



Equilibrium of forces acting on the mass at given time t satisfies the Newton's 1st Law:

$$+\downarrow \sum [-F(t) - F_s + W] = 0$$

But since we have the **dynamic force** to be: $F(t) = m \frac{d^2 y(t)}{dt^2}$

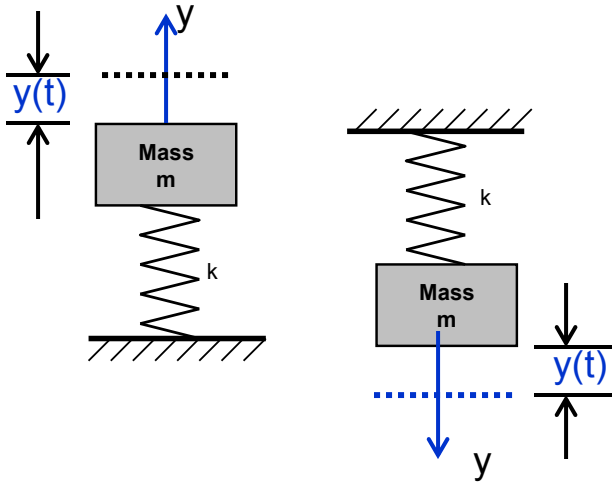
and the **spring force** to be $F_s = k[h + y(t)]$, we should have:

$$-m \frac{d^2 y(t)}{dt^2} - k[h + y(t)] + mg = 0$$

But $mg = kh$ from the static equilibrium condition, and after substituting it into the above equation, we have the following 2nd order differential equation for the instantaneous position $y(t)$ for the vibrating mass:

$$m \frac{d^2 y(t)}{dt^2} + k y(t) = 0$$

Solution of differential equation for simple mass-spring vibration



$$m \frac{d^2 y(t)}{dt^2} + k y(t) = 0 \quad (8.14)$$

where $y(t)$ = instantaneous position of the mass

Re-writing the above equation in the following form:

$$\frac{d^2 y(t)}{dt^2} + \frac{k}{m} y(t) = 0 \quad (8.14a)$$

The solution of Equation (8.14) can be obtained by comparing Equation (8.14a) with the typical 2nd order differential equation in Equation (8.1):

$$\frac{d^2 u(x)}{dx^2} + a \frac{du(x)}{dx} + bu(x) = 0 \quad (8.1)$$

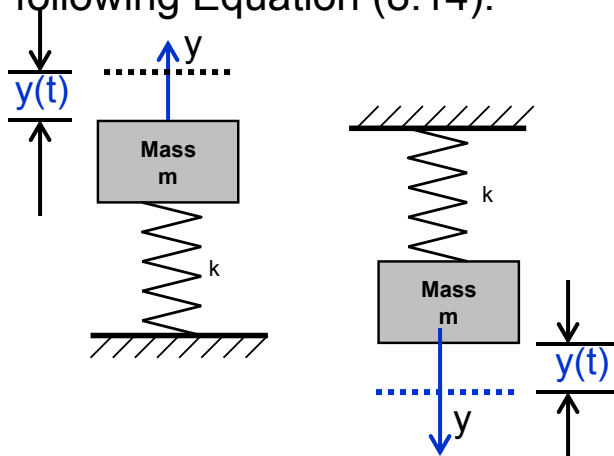
We will find that $a = 0$ and $b = k/m$ after comparing Eqs..(8.14a) and (8.1). The solution of Equation (8.1) depends on the discriminator: $a^2 - 4b$. Since k = spring constant which is a property of the spring and m = mass of the vibrating solid, the equivalent coefficient b is a +ve real number. Consequently, we will have: $a^2 - 4b = 0 - 4(k/m) < 0$, which is a Case 2 for the solution, as shown in Equation(8.8), or

$$y(t) = A \cos \sqrt{\frac{k}{m}} t + B \sin \sqrt{\frac{k}{m}} t \quad (8.15)$$

where A, B are arbitrary constants to be determined by given conditions

Physical senses of solution of differential equation (8.14) for simple mass-spring vibration:

The differential equation for the instantaneous position of the solid mass, $y(t)$ satisfies the following Equation (8.14):



$$m \frac{d^2 y(t)}{dt^2} + k y(t) = 0 \quad (8.14)$$

where $y(t)$ = instantaneous position of the mass

Upon re-writing the equation in the form:

$$\frac{d^2 y(t)}{dt^2} + \frac{k}{m} y(t) = 0 \quad (8.14a)$$

The general solution of Equation (8.14) is:

$$y(t) = c_1 \cos \omega_o t + c_2 \sin \omega_o t \quad (8.16)$$

where c_1 and c_2 are arbitrary constants to be determined by given conditions, and

$$\omega_o = \sqrt{\frac{k}{m}} \quad (8.16a)$$

is called the “*circular*”, or “*angular frequency*” of the mass-spring vibration system. Often, it represents the “*natural frequency*” of the simple mass-spring system. The unit is *rad/s*.

Corresponding to the angular frequency ω_o is the real frequency of the vibration in the following expression::

$$f = \frac{\omega_o}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (8.17)$$

Applications of Second-Order Differential Equations

Second-order linear differential equations have a variety of applications in science and engineering. In this section we explore two of them: the vibration of springs and electric circuits.

Vibrating Springs

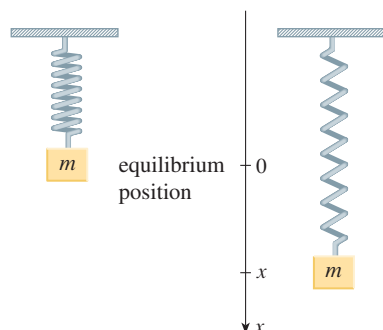


FIGURE 1

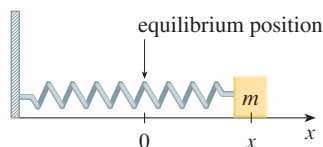


FIGURE 2

We consider the motion of an object with mass m at the end of a spring that is either vertical (as in Figure 1) or horizontal on a level surface (as in Figure 2).

In Section we discussed Hooke's Law, which says that if the spring is stretched (or compressed) x units from its natural length, then it exerts a force that is proportional to x :

$$\text{restoring force} = -kx$$

where k is a positive constant (called the **spring constant**). If we ignore any external resisting forces (due to air resistance or friction) then, by Newton's Second Law (force equals mass times acceleration), we have

$$\boxed{1} \quad m \frac{d^2x}{dt^2} = -kx \quad \text{or} \quad m \frac{d^2x}{dt^2} + kx = 0$$

This is a second-order linear differential equation. Its auxiliary equation is $mr^2 + k = 0$ with roots $r = \pm \omega i$, where $\omega = \sqrt{k/m}$. Thus, the general solution is

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

which can also be written as

$$x(t) = A \cos(\omega t + \delta)$$

where

$$\omega = \sqrt{k/m} \quad (\text{frequency})$$

$$A = \sqrt{c_1^2 + c_2^2} \quad (\text{amplitude})$$

$$\cos \delta = \frac{c_1}{A} \quad \sin \delta = -\frac{c_2}{A} \quad (\delta \text{ is the phase angle})$$

1.2 Example

A spring with a mass of 2 kg has natural length 0.5 m. A force of 25.6 N is required to maintain it stretched to a length of 0.7 m. If the spring is stretched to a length of 0.7 m and then released with initial velocity 0, find the position of the mass at any time t .

SOLUTION From Hooke's Law, the force required to stretch the spring is

$$\begin{aligned} \text{here length of the spring is } 0.7 - 0.5 = 0.2, \quad ks = mg, \\ k(0.2) = 25.6 \end{aligned}$$

so $k = 25.6/0.2 = 128$. Using this value of the spring constant k , together with $m = 2$ in Equation 1, we have

$$2 \frac{d^2x}{dt^2} + 128x = 0$$

As in the earlier general discussion, the solution of this equation is

$$\boxed{2} \quad x(t) = c_1 \cos 8t + c_2 \sin 8t$$

We are given the initial condition that $x(0) = 0.2$. But, from Equation 2, $x(0) = c_1$. Therefore, $c_1 = 0.2$. Differentiating Equation 2, we get

$$x'(t) = -8c_1 \sin 8t + 8c_2 \cos 8t$$

Since the initial velocity is given as $x'(0) = 0$, we have $c_2 = 0$ and so the solution is

$$x(t) = \frac{1}{5} \cos 8t$$

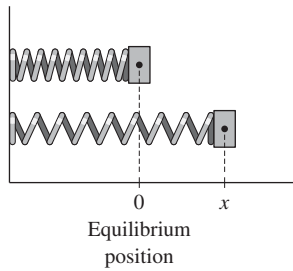


FIGURE 3 Mass m oscillating at the end of a spring

A typical example of simple harmonic motion is the motion of a mass m attached to the end of the spring. Let $x(t)$ be the position of the mass at time t , where $x = 0$ is the equilibrium position (Figure 1). Hooke's Law, which is valid for small displacements from equilibrium, states that the spring exerts a restoring force $F(x) = -kx$ where $k > 0$ is the spring constant (in units of force per unit distance). The velocity and acceleration of the mass are

$$v(t) = x'(t), \quad a(t) = x''(t)$$

Newton's law, $F = ma$ yields $-kx(t) = mx''(t)$ or

$$mx''(t) + kx(t) = 0$$

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This is the differential equation for simple harmonic motion. The characteristic equation $m\lambda^2 + k = 0$ has two purely imaginary roots $\pm\omega_0 i$, where

$$\omega_0 = \sqrt{\frac{k}{m}}$$

The general solution of (1) may be written in two ways:

$$x(t) = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t \quad \text{or} \quad x(t) = C \cos(\omega_0 t - \theta)$$

2

where $C = \sqrt{c_1^2 + c_2^2}$ and $\tan \theta = c_2/c_1$.

The maximum value of $|x(t)|$, called the **amplitude**, is equal to C . The constant ω_0 (in radians per unit time) is called the **natural angular frequency** of the oscillator. The **frequency** of the oscillations (in cycles per second) is $f = \omega_0/(2\pi)$ and the time required to complete one cycle, called the **period**, is equal to $T = 2\pi/\omega_0$.

- **Natural Angular frequency:** $\omega_0 = \sqrt{\frac{k}{m}}$
- **Frequency:** $f = \omega_0/(2\pi)$
- **Amplitude:** $C = \sqrt{c_1^2 + c_2^2}$
- **Period:** $T = \frac{2\pi}{\omega_0}$

1.3. EXAMPLE Simple Harmonic Motion A 1.5 kg mass is attached to a spring with spring constant $k = 50$ N/m. The mass is stretched 0.5 meters past equilibrium and

(a) Determine the motion of the spring.

(b) Write the motion in the form $x(t) = A \cos(\omega_0 t - \theta)$.

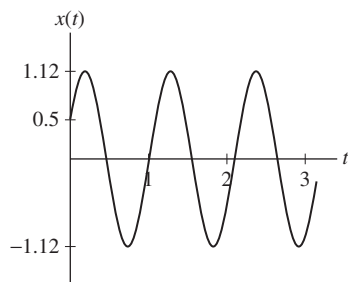


FIGURE 2 Simple Harmonic Motion of the Spring in Example 1:
 $x(t) = 1.12 \cos(5.77t - 1.11)$

Solution

(a) The position x of the mass satisfies Eq. (1) with $m = 1.5$ and $k = 50$:

$$1.5x'' + 50x = 0$$

The angular frequency is

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{50}{1.5}} \approx 5.77$$

By Eq. (2), the general solution is

$$x(t) = c_1 \cos 5.77t + c_2 \sin 5.77t$$

We use the initial conditions to determine the constants c_1 and c_2 :

$$x(0) = c_1 \cos 0 + c_2 \sin 0 = c_1 = 0.5$$

$$x'(0) = -c_1 \sin 0 + c_2 \cos 0 = c_2 = 1$$

Therefore

$$x(t) = 0.5 \cos 5.77t + \sin 5.77t$$

(b) As shown in Section 1, we can rewrite the solution as $x(t) = C \cos(\omega_0 t - \theta)$, where

$$C = \sqrt{c_1^2 + c_2^2} = \sqrt{.5^2 + 1^2} \approx 1.12, \quad \tan \theta = \frac{c_2}{c_1} = 1/.5 = 2$$

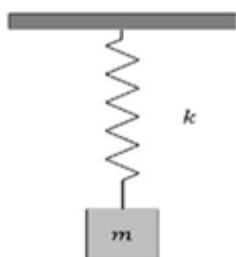
Since $c_1 > 0$, we choose θ in the interval $[0, \frac{\pi}{2})$. Thus $\theta = \tan^{-1} 2 \approx 1.11$ and (Figure 2)

$$x(t) = 1.12 \cos(5.77t - 1.11)$$

1.4. Example:

. A mass weighing 2lb stretches a spring 2 ft, the equation of the spring is released from a point 2ft below the equilibrium position with an upward velocity of 8 ft/sec.

The general equation of motion for the system is given by $m \frac{d^2x}{dt^2} + kx = 0$, where m is the mass, k is the spring constant, and x is the displacement from the equilibrium position



Hint : 1 inch=0.0833 ft (or) 1/12 ft, $g=32$ ft/sec

Sol: From $W=mg$, we have $2=m(32)$, $m=2/32=1/16$,

Equilibrium position:

$mg=ks$, $2=k(2)=1$, Therefore $k=1$,

The D.E is $mx''(t) + kx(t) = 0$

$\frac{1}{16}x''(t) + x(t) = 0$, $x''(t) + 16x(t) = 0$, The A.E is $m^2 + 16 = 0$, $m = \pm 4i$

$X(t) = A\cos 4t + B\sin 4t$

Initial conditions are $x(0)=2$ and $x'(0) = -8$

$x(0)=A$, $A=2$

$x'(t) = -4A\sin 4t + 4B\cos 4t$, when $x'(0) = -8$, $x'(0) = -4A\sin 4(0) + 4B\cos 4(0)$

$$x'(0) = 0 + 4B$$

$-8=4B$, $B= -2$, Therefore the position function is $X(t) = 2\cos 4t - 2\sin 4t$

1.5. Discussion Questions:

1. The mass-spring system is displaced from its equilibrium position, and the resulting oscillations are observed and recorded in engineering laboratories. A mass weighing 12lb stretches a spring 2 ft, the equation of the spring is released from a point 1foot below the equilibrium position with an upward velocity of 4 ft/sec. The general equation of motion for the system is given by $m \frac{d^2 x}{dt^2} + kt = 0$, where m is the mass, k is the spring constant, and x is the displacement from the equilibrium position. Compute the position function.
2. The mass-spring system is displaced from its equilibrium position, and the resulting oscillations are observed and recorded in engineering laboratories. A mass weighing 2lb stretches a spring 6 inch, the equation of the spring is released from a point 8inch below the equilibrium position with a upward velocity of $4/3$ ft/sec. The general equation of motion for the system is given by $m \frac{d^2 x}{dt^2} + kt = 0$, where m is the mass, k is the spring constant, and x is the displacement from the equilibrium position. Compute the position function.

